

Lecture 11

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1 Packing common bases

Let $M_1 = (E, \mathcal{I}_1)$ and $M_2 = (E, \mathcal{I}_2)$ be matroids of the same rank. A *common base* is a set that is a base of both matroids. The common-base packing problem asks for pairwise disjoint common bases, or, in its partition form, for common bases $B_1, \dots, B_k$ satisfying $E = B_1 \dot{\cup} \dots \dot{\cup} B_k$.

A necessary condition for such a partition is that E can be partitioned into k bases of M_1 and also into k bases of M_2 . In general this condition is not sufficient. It becomes sufficient under a strong exchange hypothesis.

1.1 Strong base orderability

Definition 1.1 (Strong base orderability). A matroid $M = (E, \mathcal{I})$ is *strongly base-orderable* if, for every two bases B, B' , there is a bijection $\varphi : B \rightarrow B'$ such that

$$(B \setminus X) \cup \varphi(X)$$

is a base for every $X \subseteq B$. Such a bijection is called a *strong base ordering*. Equivalently, both

$$(B \setminus X) \cup \varphi(X) \quad \text{and} \quad (B' \setminus \varphi(X)) \cup X$$

are bases for every $X \subseteq B$.

The equivalence follows by applying the first condition to $B \setminus X$. Partition matroids are strongly base-orderable. The notion goes back to Brualdi and Scrimger [3].

Not every graphic matroid has this property. For example, in $M(K_4)$ consider the spanning trees

$$B = \{12, 13, 24\}, \quad B' = \{14, 23, 34\}.$$

In a symmetric exchange between these bases, each of the edges 13 and 24 can be paired only with 34. Hence no strong base ordering exists. See also Bonin and Savitsky [2] for the excluded-minor theory of strong base orderability.

Theorem 1.2 (Davies–McDiarmid [5]). *Let M_1 and M_2 be strongly base-orderable matroids on E . If E can be partitioned into k bases of M_1 and into k bases of M_2 , then E can be partitioned into k common bases.*

Proof. Choose partitions

$$E = X_1 \dot{\cup} \dots \dot{\cup} X_k = Y_1 \dot{\cup} \dots \dot{\cup} Y_k,$$

where every X_i is a base of M_1 and every Y_i is a base of M_2 . Among all such pairs of partitions, choose one maximizing

$$\Phi = \sum_{i=1}^k |X_i \cap Y_i|.$$

Suppose that some off-diagonal intersection $X_i \cap Y_j$ is nonempty, where $i \neq j$. Let

$$\varphi_1 : X_i \longrightarrow X_j, \quad \varphi_2 : Y_i \longrightarrow Y_j$$

be strong base orderings in M_1 and M_2 , respectively. Form a graph whose edges are the pairs $x\varphi_1(x)$ and $y\varphi_2(y)$. It is the union of two matchings, so each component is a path or an even cycle; in particular, it is bipartite. Let (S, T) be a bipartition.

Set

$$\begin{aligned} X'_i &= (X_i \cup X_j) \cap S, & X'_j &= (X_i \cup X_j) \cap T, \\ Y'_i &= (Y_i \cup Y_j) \cap S, & Y'_j &= (Y_i \cup Y_j) \cap T. \end{aligned}$$

Every edge of either matching has one endpoint in each color class. Strong base orderability therefore implies that X'_i, X'_j are bases of M_1 and that Y'_i, Y'_j are bases of M_2 .

Every element of $(X_i \cup X_j) \cap (Y_i \cup Y_j)$ now lies in the same indexed X -block and Y -block: an element in S lies in both primed i -blocks, and an element in T lies in both primed j -blocks. Hence

$$|X'_i \cap Y'_i| + |X'_j \cap Y'_j| = |(X_i \cup X_j) \cap (Y_i \cup Y_j)|.$$

This contains all old diagonal intersections and, because $X_i \cap Y_j \neq \emptyset$, at least one additional element. Thus Φ strictly increases, a contradiction. There are therefore no off-diagonal intersections, and the two partitions coincide: $X_i = Y_i$ for every i . $\square$

The theorem is structural. To turn this proof into an algorithm, one also needs an efficient way to produce the strong base orderings.

1.2 Two motivating examples

Bipartite matching. Let $G = (L \cup R, E)$ be a k -regular bipartite multigraph. On E , take the partition matroid that allows at most one edge incident to each vertex of L , and the analogous partition matroid for R . Their common bases are perfect matchings. Since partition matroids are strongly base-orderable, the theorem recovers König's decomposition of a k -regular bipartite graph into k perfect matchings.

Rooted arborescences. Let $D = (V, A)$ be a directed graph with root r . A spanning arborescence rooted at r is a common base of

- the graphic matroid obtained by forgetting the orientations, and
- the partition matroid allowing no arc entering r and at most one arc entering each vertex of $V \setminus \{r\}$.

For $X \subseteq V$, let $\delta^-(X)$ be the arcs entering X .

Theorem 1.3 (Edmonds's disjoint-branchings theorem [6]). *A directed graph contains k pairwise arc-disjoint spanning arborescences rooted at r if and only if*

$$|\delta^-(X)| \geq k \quad \text{for every nonempty } X \subseteq V \setminus \{r\}.$$

The graphic matroid of K_4 is not strongly base-orderable, so this theorem is not a consequence of the Davies–McDiarmid theorem. It is a different common-base packing theorem with a cut characterization.

2 Submodular flows

Let $D = (V, A)$ be a directed graph. For $U \subseteq V$, write $\delta^+(U)$ and $\delta^-(U)$ for the arcs leaving and entering U , respectively, and define the signed cut vector $d_U \in \{-1, 0, 1\}^A$ by

$$d_U^\top x = x(\delta^+(U)) - x(\delta^-(U)).$$

Definition 2.1 (Crossing family and crossing submodularity). Two sets $X, Y \subseteq V$ *cross* if all four sets

$$X \cap Y, \quad X \setminus Y, \quad Y \setminus X, \quad V \setminus (X \cup Y)$$

are nonempty. A family $\mathcal{C} \subseteq 2^V$ is *crossing* if $X \cap Y, X \cup Y \in \mathcal{C}$ whenever $X, Y \in \mathcal{C}$ cross. A function $f : \mathcal{C} \rightarrow \mathbb{R}$ is *crossing submodular* if, for every crossing pair,

$$f(X) + f(Y) \geq f(X \cap Y) + f(X \cup Y).$$

The associated submodular-flow polyhedron is

$$P(f) = \{x \in \mathbb{R}^A : d_U^\top x \leq f(U) \text{ for every } U \in \mathcal{C}\}.$$

Definition 2.2 (Box-TDI). A rational system $Q = \{x : Ax \leq b\}$ is *box-TDI* if, after adding arbitrary rational bounds $a \leq x \leq b'$, the resulting system is totally dual integral: for every integral objective with a finite optimum, the dual has an integral optimum.

Theorem 2.3 (Edmonds–Giles [7]). *If $f : \mathcal{C} \rightarrow \mathbb{Q}$ is crossing submodular, then the defining system of $P(f)$ is box-TDI. Consequently, if f and the lower and upper bounds are integral, then*

$$P(f) \cap \{x : a \leq x \leq b\}$$

is an integral polyhedron.

2.1 The primal–dual pair and uncrossing

With lower bounds a and upper bounds b , the primal problem is

$$\begin{aligned} & \text{maximize} && w^\top x \\ & \text{subject to} && d_U^\top x \leq f(U) \quad (U \in \mathcal{C}), \\ & && a \leq x \leq b. \end{aligned} \tag{P}$$

Its dual is

$$\begin{aligned} & \text{minimize} && \sum_{U \in \mathcal{C}} f(U) y_U + \sum_{e \in A} b_e \beta_e - \sum_{e \in A} a_e \alpha_e \\ & \text{subject to} && \sum_{U: e \in \delta^+(U)} y_U - \sum_{U: e \in \delta^-(U)} y_U + \beta_e - \alpha_e = w_e \quad (e \in A), \\ & && y, \alpha, \beta \geq 0. \end{aligned} \tag{D}$$

Here is the core of the box-TDI proof. Among all optimal dual solutions, choose one minimizing

$$\Psi(y) = \sum_{U \in \mathcal{C}} y_U |U| |V \setminus U|.$$

Suppose $X, Y \in \text{supp}(y)$ cross, and put $\varepsilon = \min\{y_X, y_Y\}$. Replace

$$y_X, y_Y \quad \text{by} \quad y_X - \varepsilon, y_Y - \varepsilon$$

and add ε to each of $y_{X \cap Y}$ and $y_{X \cup Y}$. The identity

$$d_X + d_Y = d_{X \cap Y} + d_{X \cup Y}$$

preserves every dual equality, while crossing submodularity does not increase the dual objective. Since the original solution was optimal, the new one is optimal as well. But

$$\begin{aligned} & |X| |V \setminus X| + |Y| |V \setminus Y| \\ & - |X \cap Y| |V \setminus (X \cap Y)| - |X \cup Y| |V \setminus (X \cup Y)| = 2|X \setminus Y| |Y \setminus X| > 0, \end{aligned}$$

contradicting the choice of y . Thus $\text{supp}(y)$ is *cross-free*: no two members cross.

The signed cut matrix $D_{\mathcal{L}}$ whose rows are d_U for a cross-free family $\mathcal{L}$ is totally unimodular [13]. Hence its transpose is totally unimodular, and so is the dual constraint matrix

$$[D_{\mathcal{L}}^T \quad I \quad -I].$$

Therefore, when w is integral, the dual has an integral optimal vertex. This proves box-TDI. Notice that a cross-free family need not literally be laminar: two members may have union V . After complementing the sets that contain a fixed root, one obtains the usual laminar picture.

3 Equivalence with polymatroid intersection

Submodular flow and polymatroid intersection are two forms of the same optimization framework. We record both reductions; see Frank and Tardos [9] for the general theory.

3.1 From polymatroid intersection to submodular flow

Let $f_1, f_2 : 2^E \rightarrow \mathbb{R}_+$ be normalized, nondecreasing submodular functions, and let

$$P(f_i) = \{x \in \mathbb{R}_+^E : x(S) \leq f_i(S) \text{ for all } S \subseteq E\}.$$

Make two copies E' and E'' of E , and for every $e \in E$ add an arc $a_e : e' \rightarrow e''$ carrying x_e . Use the crossing family

$$\mathcal{C} = \{S' : S' \subseteq E'\} \cup \{E' \cup S'' : S'' \subseteq E''\}.$$

Define

$$h(S') = f_1(S'), \quad h(E' \cup S'') = f_2(E \setminus S'').$$

The set E' has two representations; there we use $h(E') = \min\{f_1(E), f_2(E)\}$. The resulting h is crossing submodular. Moreover,

$$d_{S'}^T x = x(S'), \quad d_{E' \cup S''}^T x = x(E \setminus S'').$$

Thus, together with $x \geq 0$, the submodular-flow inequalities are exactly the inequalities defining $P(f_1) \cap P(f_2)$.

3.2 From submodular flow to polymatroid intersection

First assume $0 \leq x_e \leq 1$ for every arc $e = (u, v)$. Replace e by two incidence elements e^+ and e^- , assign e^+ to its tail u and e^- to its head v , and set

$$z_{e^+} = x_e, \quad z_{e^-} = 1 - x_e.$$

Together with $z \geq 0$, the equations $z_{e^+} + z_{e^-} = 1$ describe the base polytope of the direct sum of one rank-one matroid on each pair $\{e^+, e^-\}$. To make the encoding injective even at isolated vertices, add a tagged loop q_v of fixed coordinate zero for every $v \in V$. For $U \subseteq V$, let $H(U)$ be the set of incidence elements and tagged loops assigned to vertices of U . A direct count gives

$$z(H(U)) = |A[U]| + |\delta^-(U)| + x(\delta^+(U)) - x(\delta^-(U)).$$

Consequently, if

$$g(H(U)) = |A[U]| + |\delta^-(U)| + f(U),$$

then

$$z(H(U)) \leq g(H(U)) \iff d_U^\top x \leq f(U).$$

The modular incidence terms preserve crossing submodularity. The standard extension and Dilworth-truncation construction converts these inequalities into a polymatroid without changing the feasible points on the partition-matroid base face. General rational lower and upper bounds follow by scaling and the standard capacity transformation. This sketches the polymatroid-intersection formulation of submodular flow; the full extension argument is given in [9].

4 Dicuts and dijoins

Definition 4.1. A *dicut* is a directed cut $\delta^+(U)$ with $\emptyset \neq U \subsetneq V$ and $\delta^-(U) = \emptyset$. A *dijoin* is a set of arcs that meets every nonempty dicut.

Theorem 4.2 (Lucchesi–Younger [12]). *Let $w \in \mathbb{Z}_+^A$, and let $\mathcal{D}$ be the family of nonempty dicuts. Then*

$$\min\{w(J) : J \text{ is a dijoin}\} = \max \left\{ \sum_{C \in \mathcal{D}} y_C : \sum_{C \ni e} y_C \leq w_e \ (e \in A), \quad y \in \mathbb{Z}_+^{\mathcal{D}} \right\}.$$

In particular, the minimum cardinality of a dijoin equals the maximum number of pairwise arc-disjoint dicuts.

This is a directed analogue of the max-flow min-cut theorem. One proof is obtained by expressing the covering problem as a submodular flow and using the box-TDI theorem.

4.1 Clutters, blockers, and Woodall’s conjecture

A *clutter* is a family $\mathcal{F} \subseteq 2^A$ in which no member contains another. Its covering polyhedron and blocker are

$$Q(\mathcal{F}) = \{x \in \mathbb{R}_+^A : x(F) \geq 1 \text{ for all } F \in \mathcal{F}\}, \quad b(\mathcal{F}) = \{\text{inclusion-minimal transversals of } \mathcal{F}\}.$$

The clutter is *ideal* if $Q(\mathcal{F})$ is integral.

Let $\mathcal{F}$ be the clutter of inclusion-minimal dicuts. Its blocker consists of the inclusion-minimal dijoins. Lucchesi–Younger implies that $\mathcal{F}$ is ideal. Lehman’s blocker theorem [11] says

$$\mathcal{F} \text{ is ideal} \iff b(\mathcal{F}) \text{ is ideal.}$$

Thus the covering polyhedron of minimal dijoins is also integral. This ideality statement does *not* by itself imply an integral packing theorem for dijoins.

Conjecture 4.3 (Woodall [15]). If every nonempty dicut of a directed graph has at least τ arcs, then the graph contains τ pairwise arc-disjoint dijoins. Equivalently,

$$\max\{\text{number of pairwise disjoint dijoins}\} = \min\{|C| : C \text{ is a nonempty dicut}\}.$$

Frank and Tardos [8] showed that Woodall’s conjecture can be encoded as a common-base packing problem on an extended ground set. This does not mean that dijoins themselves are literally common bases on the original arc set.

The following is the best constant-factor guarantee presently needed here.

Theorem 4.4 (Cornuéjols–Liu–Ravi [4]). *If the minimum size of a nonempty dicut is τ , then one can find*

$$\left\lfloor \frac{\tau}{6} \right\rfloor$$

pairwise arc-disjoint dijoins in polynomial time.

This gives an asymptotic $1/6$ -approximation to Woodall’s predicted bound. The proof uses, among other ingredients, Seymour’s nowhere-zero 6-flow theorem [14].

5 An oracle lower bound for packing two common bases

Even deciding whether two disjoint common bases exist is exponentially hard in the independence-oracle model. Bérczi and Schwarcz established this oracle lower bound in general [1]. The sparse-paving construction below is the simpler information-theoretic argument highlighted by Hörsch, Imolay, Mizutani, Oki, and Schwarcz [10]; through non-zero common bases over $\mathbb{Z}_2$, their argument also gives a new proof of the common-base packing lower bound.

Lemma 5.1 (Sparse-paving construction). *Let $\mathcal{H} \subseteq \binom{E}{r}$ satisfy $|H \cap H'| \leq r - 2$ for all distinct $H, H' \in \mathcal{H}$. Then*

$$\mathcal{B} = \binom{E}{r} \setminus \mathcal{H}$$

is the set of bases of a rank- r matroid, provided $\mathcal{B} \neq \emptyset$.

Proof. It is enough to verify basis exchange. Let $X, Y \in \mathcal{B}$ and $e \in X \setminus Y$. If $|Y \setminus X| = 1$, then exchanging the unique element produces Y , which is a base. Otherwise, among the sets $X - e + f$ with $f \in Y \setminus X$, at most one can lie in $\mathcal{H}$: any two such sets intersect in $r - 1$ elements, whereas distinct members of $\mathcal{H}$ intersect in at most $r - 2$. Since $|Y \setminus X| \geq 2$, at least one exchange produces a member of $\mathcal{B}$. $\square$

Theorem 5.2. *For a ground set of size $n = 2r$, where r is odd, any deterministic independence-oracle algorithm deciding whether two matroids have two disjoint common bases requires at least*

$$2^{r-1} = 2^{n/2-1}$$

oracle queries in the worst case.

Proof. Let

$$E = A \dot{\cup} B, \quad A = \{a_1, \dots, a_r\}, \quad B = \{b_1, \dots, b_r\},$$

and pair the elements as $P_i = \{a_i, b_i\}$. Let M_1 be the rank- r partition matroid allowing at most one element from each pair. Its bases are precisely the transversals of the pairs.

Let

$$\mathcal{H} = \{F \subseteq E : |F \cap P_i| = 1 \text{ for every } i, |F \cap A| \text{ is odd}\}.$$

Two distinct members of $\mathcal{H}$ differ on a positive even number of pairs, and therefore intersect in at most $r - 2$ elements. By the lemma,

$$\mathcal{B}_2 = \binom{E}{r} \setminus \mathcal{H}$$

is the set of bases of a rank- r matroid M_2 .

This is a *no-instance*. Indeed, if C and D are disjoint bases of M_1 , then $D = E \setminus C$. Since r is odd, exactly one of $|C \cap A|$ and $|D \cap A| = r - |C \cap A|$ is odd. Hence exactly one of C, D belongs to $\mathcal{H}$ and fails to be a base of M_2 .

For each $F \in \mathcal{H}$, define a second matroid M_2^F whose bases are

$$\binom{E}{r} \setminus (\mathcal{H} \setminus \{F\}).$$

This is a *yes-instance*: F is now a base, and its complement $E \setminus F$ is an even transversal because r is odd, so it was already a base of M_2 . Thus F and $E \setminus F$ are disjoint common bases of M_1 and M_2^F .

Every set of size less than r extends to a base: an $(r - 1)$ -set has at most one forbidden r -element extension, by the intersection property of $\mathcal{H}$, and smaller sets can first be extended to size $r - 1$. Sets larger than r are dependent in both matroids. Hence the independence oracles for M_2 and M_2^F differ only on F . Moreover, every proper superset of F contains an allowed r -set: replace any element of F by an element outside F . Thus such a superset has rank r in both matroids. There are $|\mathcal{H}| = 2^{r-1}$ choices for F . Against the no-instance oracle, a deterministic algorithm must query every member of $\mathcal{H}$; otherwise an unqueried F gives a yes-instance with exactly the same transcript. This proves the lower bound. $\square$

The construction also works with rank oracles, since the rank answers for M_2 and M_2^F differ only on F . It illustrates why the existence of clean min-max theorems for special classes does not lead to a general oracle algorithm for common-base packing.

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